

Theory Lectures on Manufacturing Process (NMES 101) CNC – Computer Numerical Control



Dr. Suman Saha

Assistant Professor

Department of Mechanical Engineering

IIT (ISM) Dhanbad

Content is prepared for theory lecture for
1st Year BTech level students for common
course on Manufacturing Process.
Lecture duration $\approx$ 45 min



Automation is the technology by which a process or task is accomplished with minimum human intervention.

Manufacturing Automation started in 1946 in the Ford Motor company.

“**Degree of automation**” defines the level of human intervention required.

A(0) = Manual machines (no automation). [Can you give example?](#)

A(5) = Computer controlled with closed loop feedback. [Example?](#)

A(9) = Super machines that command human.

Manufacturing System is defined to be a collection of integrated equipment and manufacturing resources whose function is to perform one or more processing and/or assembly operations.

- Production machine and tools
- Material handling and work positioning devices
- Computer system
- Human observer/controller.



Manufacturing Process is just one subset of entire manufacturing system





Typical conventional milling machine



Typical CNC milling machine



CNC lathe



CNC milling machine



CNC horizontal machining center



CNC vertical machining center



Five-axis CNC machining center



CNC horizontal boring and milling machine



EDM



CNC laser cutting machine



CNC plasma cutting machine



3D printer

CNC is not a separate machining process. It is one way of controlling the machine. Instead of manual control, computer systems are used to control the machines in CNC.

Virtually all machines can be controlled by CNC system. In other words, all manufacturing processes can have their CNC version. For example,

- CNC lathe
- CNC milling machine
- CNC drilling machine
- CNC Electric-discharge machine (EDM)
- CNC Electro-chemical machine (ECM)
- CNC welding machine
- CNC gas cutting machine
- CNC laser cutting/welding machine
- CNC abrasive water jet cutting machine
- CNC sheet metal forming machine
- CNC sand molding machine
- CNC roughness measurement machine, etc.







Automated factories



How these CNC machines are controlled?

By **CNC programming** (One such programming method is based on **G&M code**)



G & M codes (common command lines)

G code	Interpretation / task to be performed
G00	Rapid Move
G01	Linear Feed Move
G02	Clockwise Arc Feed Move
G03	Counter Clockwise Arc Feed Move
G21	Millimetre unit for dimensioning
G52	Local Coordinate System
G90	Absolute Position Mode

M code	Interpretation / task to be performed
M00	Program stop
M03	Tool rotation start (clockwise - CW)
M04	Tool rotation start (Counter Clockwise - CCW)
M06	Tool change (give tool number to be fixed now)
M08	Coolant / cutting fluid supply on
M09	Coolant / cutting fluid supply off
M30	End of program reset

N indicates the beginning of a new line in code

% (Percentage symbol) indicates **comment** that will be neglected by interpreter

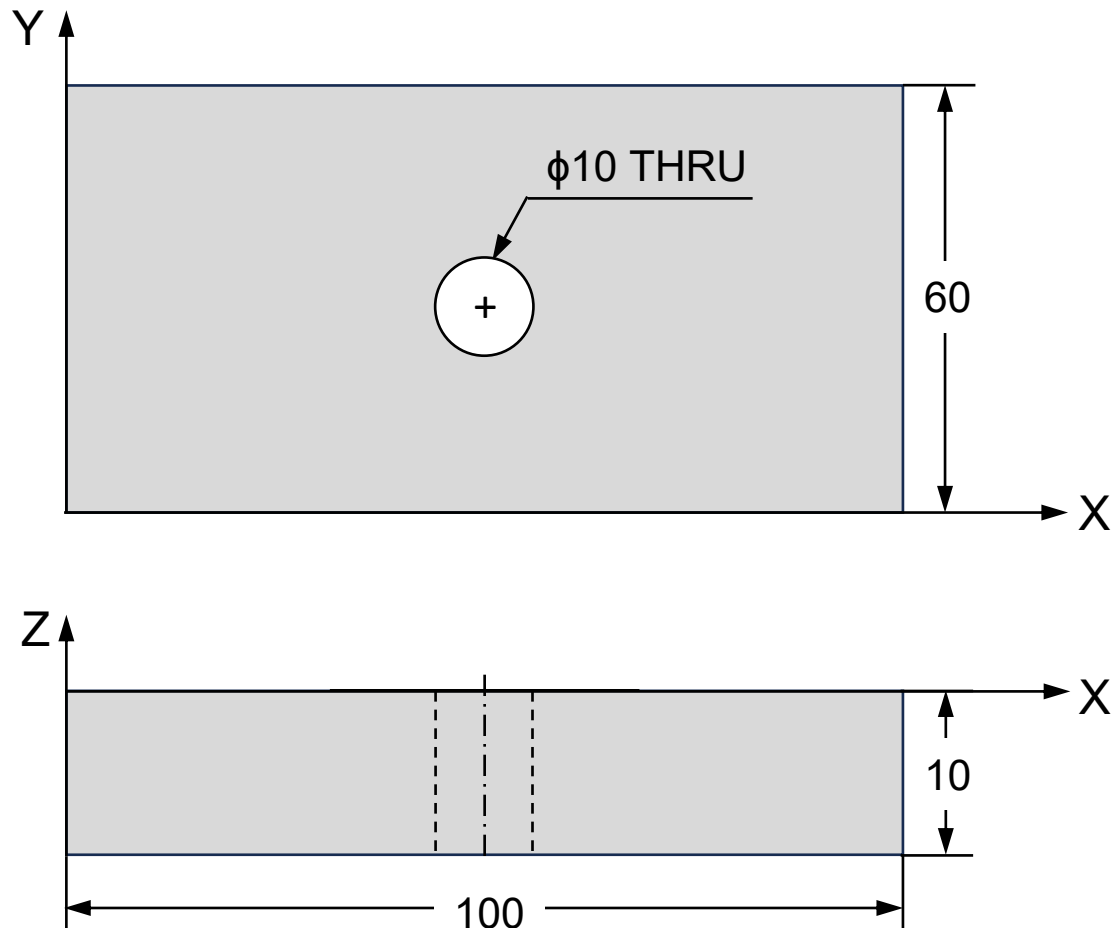
; (semicolon symbol) indicates **end of a line** (commands are executed line after line)

S symbol indicates the rotational speed value.

F symbol indicates the feed value.



Question: Write a G&M code program for drilling a 10 mm diameter through hole located at the center of 100×60×10 mm³ rectangular plate. Available drill (T05) has diameter of 10 mm.



% Slot milling CNC program

```
N01 G21 G52 G90; % mm unit, local coordinate, absolute mode
```

```
N05 M06 T05; % Change tool and attach the drill no. 05
```

```
N10 M03 S350; % Start tool rotation at 350 rpm in CW direction
```

```
N15 G00 X0 Y0 Z100; % Rapidly move to (0, 0, 100) point
```

```
N20 G00 X50 Y30 Z02; % Come to drill center but at small height
```

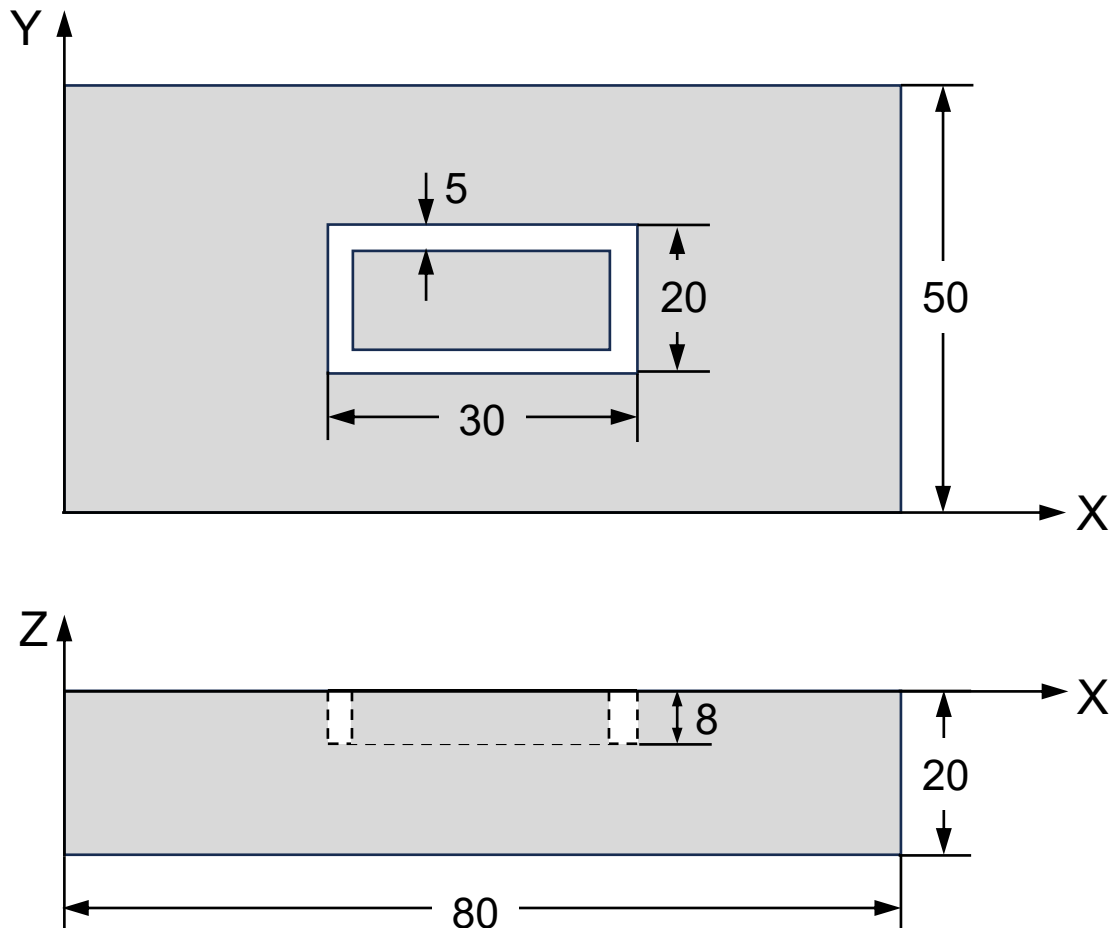
```
N25 G01 Z-12 F12 M08; % Give depth at linear feed of 12 mm/min
```

```
N30 G01 Z02; % Return back vertically above top surface
```

```
N50 G00 Z100 M09; % Fast retract to a safe location
```

```
N55 M30; % End of program
```

Question: Write a G&M code program for milling a 5-mm-wide rectangular slot located at the center of 80×50×20 mm³ rectangular plate. Available end-milling tool (T012) diameter is 5 mm.

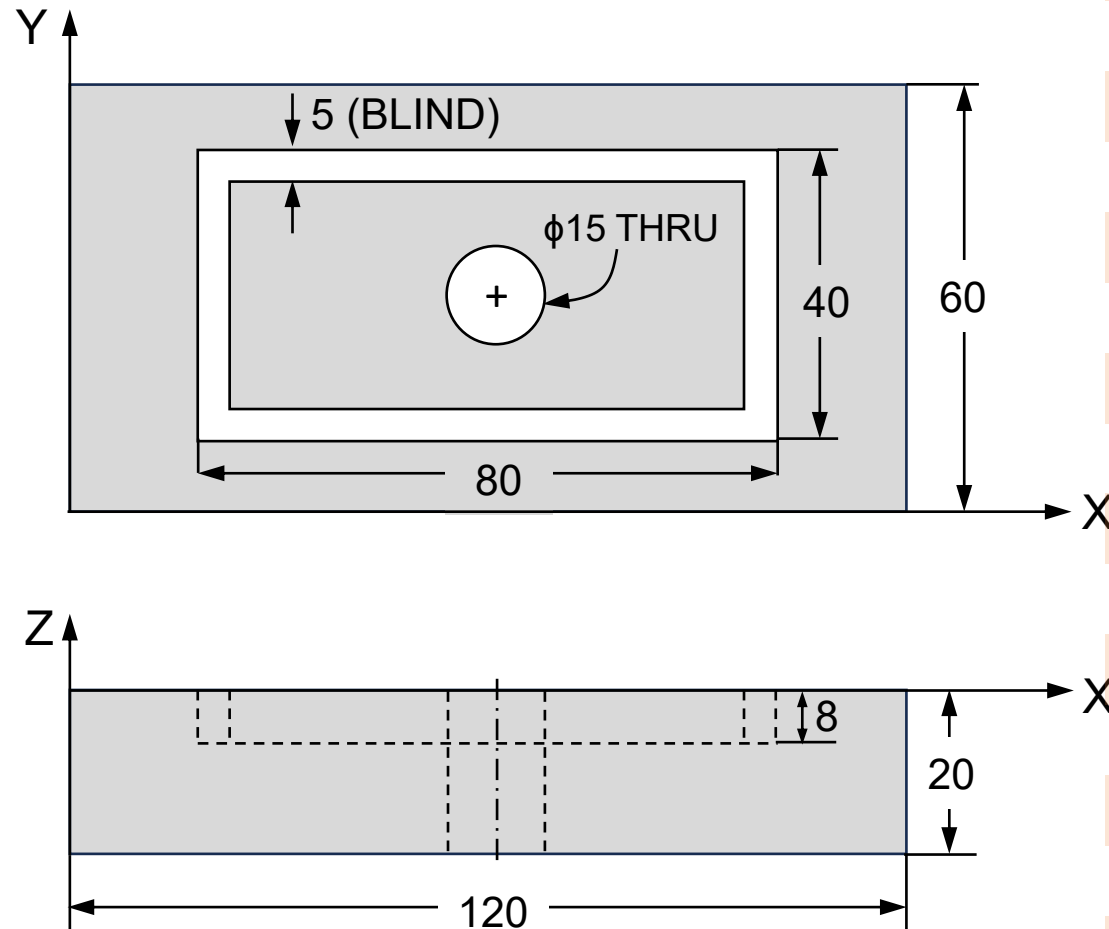


% Slot milling CNC program

```

N01 G21 G52 G90; % mm unit, local coordinate, absolute mode
N05 M06 T012; % Change tool and attach the example tool no. 12
N10 M03 S600; % Start tool rotation at 600 rpm in CW direction
N15 G00 X0 Y0 Z100; % Rapidly move to (0, 0, 100) point
N20 G00 X27.5 Y17.5 Z02; % Come to pocket corner (tool radius)
N25 G01 Z-08 F30 M08; % Give depth at linear feed of 30 m/min
N30 G01 Y32.5; % Move in Y direction to top-left corner of pocket
N35 G01 X52.5; % Move in X direction to top-right corner
N40 G01 Y17.5; % Move in Y direction to bottom-right corner
N45 G01 X27.5; % Move in X direction to bottom-left corner
N50 G01 Z2; % Bring tool out of the workpiece
N55 G00 Z100 M09; % Retract the tool vertically to a safe location
N60 M30; % End of program
  
```


Question: Write a G&M code program for milling a 5-mm-wide blind rectangular slot located at the center of 80×50×20 mm³ rectangular plate. There is also a through hole located at the center of the plate. Available end-milling tool (T11) diameter is 5 mm and a drill (T04) of diameter 15 mm.



N01 G21 G52 G90; % mm unit, local coordinate, absolute mode

N05 M06 T11; % Change tool and attach the end-milling tool

N10 M03 S600; % Start tool rotation at 600 rpm in CW direction

N15 G00 X0 Y0 Z100; % Rapidly move to (0, 0, 100) point

N20 G00 X22.5 Y12.5 Z02; % Come to bottom left pocket corner

N25 G01 F30 Z-08 M08; % Give depth at a linear feed of 30 m/min

N30 Y47.5; % Move in Y direction to top-left corner of pocket

N35 X97.5; % Move in X direction to top-right corner

N40 Y12.5; % Move in Y direction to bottom-right corner

N45 X22.5; % Move in X direction to bottom-left corner

N50 G01 Z2; % Bring tool out of the workpiece

N55 G00 Z100 M09 M00; % Retract the tool to a safe location

N60 M06 T04; % Attach the drill tool now

N65 M03 S300; % Start tool rotation at 300 rpm in CW direction

N70 G00 X0 Y0 Z100; % Rapidly move to (0, 0, 100) point

N75 X60 Y30 Z02; % Come to drill center

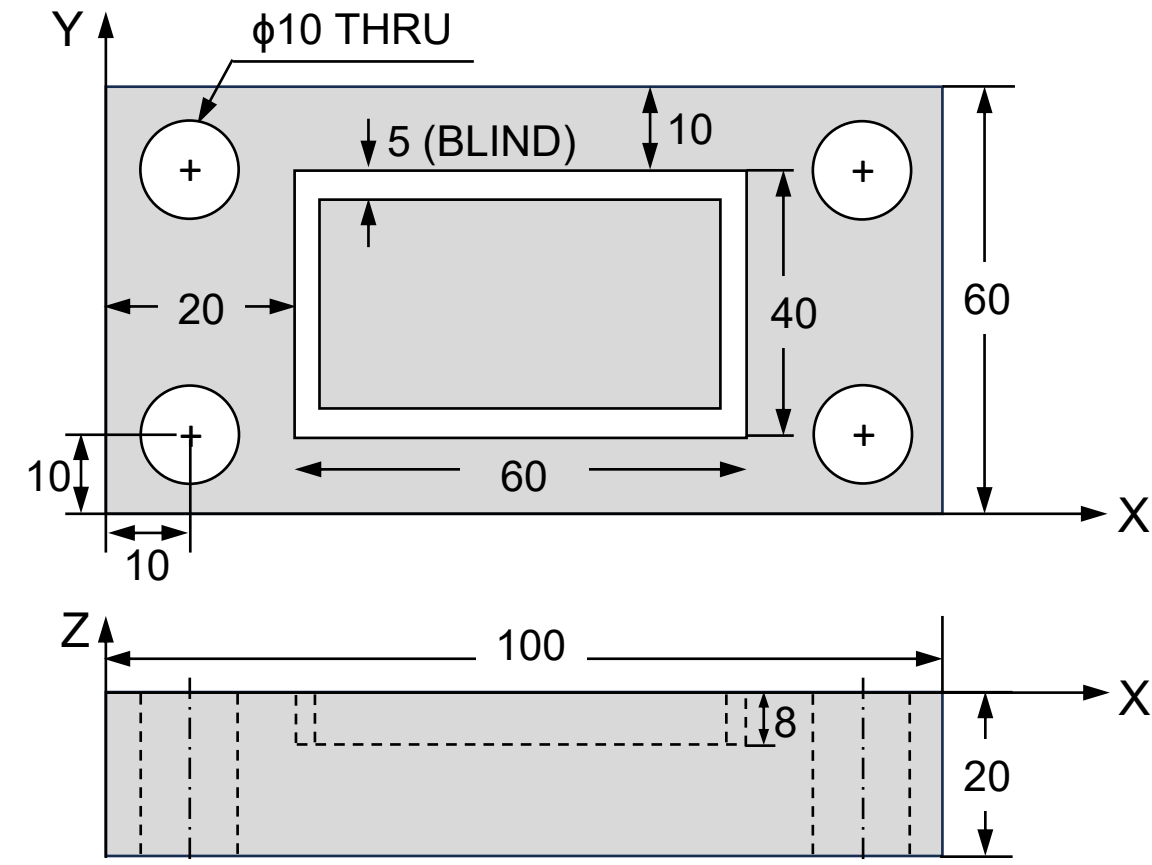
N80 G01 F15 Z-25 M08; % Vertically downward movement for drilling

N85 G01 Z2; % Take out the tool from hole

N90 G00 Z100 M09; % Retract the tool to a safe location

N950 M30; % End of program

Question: Write a G&M code program for cutting a rectangular blind slot and four symmetric through holes on a $100 \times 60 \times 20$ mm³ rectangular plate, as shown in the adjacent figure. Available end-milling tool (T02) has diameter of 10 mm and drill (T06) has diameter of 10 mm.



N01 G21 G52 G90; % mm unit, local coordinate, absolute mode

N05 M06 T06; % Change tool and attach the end-milling tool

N10 M03 S600; % Start tool rotation at 600 rpm in CW direction

N15 G00 X0 Y0 Z100; % Rapidly move to (0, 0, 100) point

N20 G00 X22.5 Y12.5 Z02; % Come to bottom left pocket corner

N25 G01 F30 Z-08 M08; % Give depth at a linear feed of 30 m/min

N30 Y47.5; % Move in Y direction to top-left corner of pocket

N35 X77.5; % Move in X direction to top-right corner

N40 Y12.5; % Move in Y direction to bottom-right corner

N45 X22.5; % Move in X direction to bottom-left corner

N50 G01 Z2; % Bring tool out of the workpiece

N55 G00 Z100 M09; % Retract the tool to a safe location

N60 M00; % Intermediate program stop

N65 M06 T04; % Attach the drill tool now

N70 M03 S300; % Start tool rotation at 300 rpm in CW direction

N75 G00 X0 Y0 Z100; % Rapidly move to safe (0, 0, 100) point

N80 X10 Y10 Z02; % Come to left-bottom hole center

N85 G01 F15 Z-25 M08; % Vertically downward movement for drilling

N90 G01 Z2; % Take out the tool from hole

N95 X10 Y50 Z02; % Come to left-bottom hole center

After previous page

N100 G01 F15 Z-25 M08; % Vertically downward movement for drilling

N105 G01 Z2; % Take out the tool from hole

N110 X90 Y50 Z02; % Come to left-bottom hole center

N115 G01 F15 Z-25 M08; % Vertically downward movement for drilling

N120 G01 Z2; % Take out the tool from hole

N125 X90 Y10 Z02; % Come to left-bottom hole center

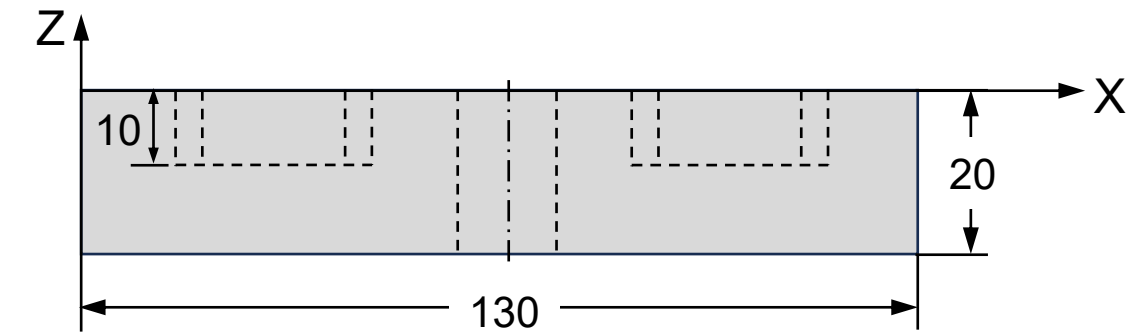
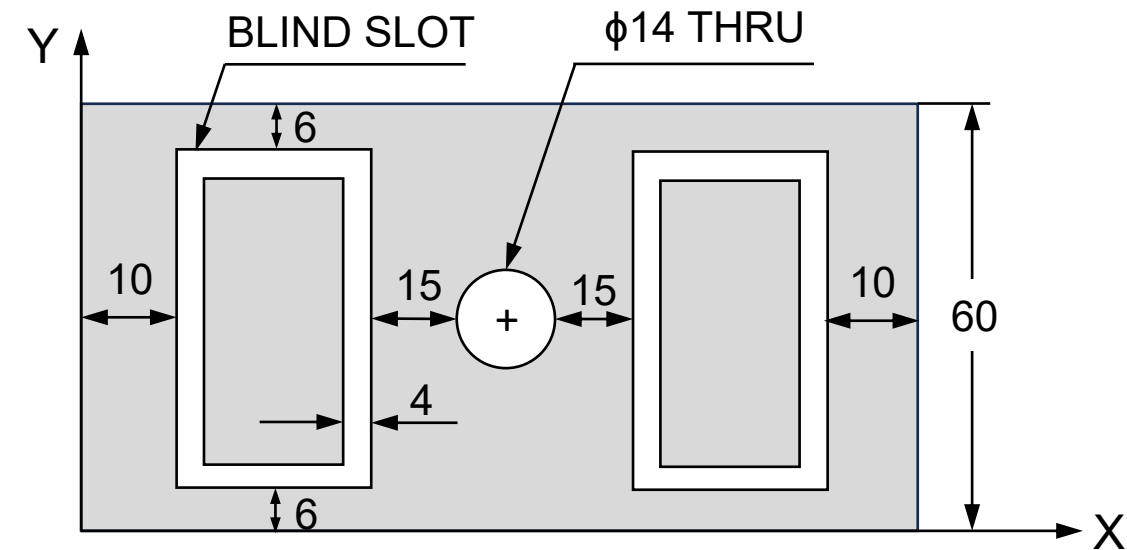
N130 G01 F15 Z-25 M08; % Vertically downward movement for drilling

N135 G01 Z2; % Take out the tool from hole

N140 G00 X0 Y0 Z100; % Move to the safe location

N145 M30; Program stop

Question: Write a CNC G&M code program for cutting a two identical rectangular blind slots and one through hole on a $130 \times 60 \times 20$ mm³ rectangular plate, as shown in the adjacent figure. Available end-milling tool (T05) has diameter of 4 mm and drill (T02) has diameter of 14 mm.



N125 G01 F30 Z02; % Vertically retract the tool outside the workpiece

N130 G00 X0 Y0 Z300 M09; % Move to the safe location

N135 M30; % Program stop

N01 G21 G52 G90; % mm unit, local coordinate, absolute mode

N05 M06 T02; % Change tool and attach the drilling tool

N10 M03 S200; % Start tool rotation at 200 rpm in CW direction

N15 G00 X0 Y0 Z300; % Rapidly move to the safe point (0, 0, 300)

N20 G00 X65 Y30 Z01; % Come to hole center without touching

N25 G01 F30 Z-27 M08; % Give depth at a linear feed of 30 m/min

N30 G01 F30 Z02; % Come out of the hole

N35 G00 X0 Y0 Z300 M09; % Move to safe location before tool change

N40 M00; % Intermediate program stop

N45 M06 T05; % Attach the end-milling tool now

N50 M03 S800; % Start tool rotation at 800 rpm in CW direction

N55 G00 X0 Y0 Z300; % Rapidly move to the safe point (0, 0, 300)

N60 G00 X12 Y8 Z1; % Rapidly come at the bottom-left corner of slot

N65 G01 F30 X12 Y8 Z-10 M08; % Give depth there to start milling

N70 G01 F30 Y52; % Move to the top-left corner of slot

N75 G01 F30 X41; % Move to the top-right corner of slot

N80 G01 F30 Y8; % Move to the bottom-right corner of slot

N85 G01 F30 X12; % Move back to the bottom-left corner of slot

N90 G01 F30 Z02; % Take the tool out of the slot

N95 G00 X89 Y8 Z02; % Come to the left-bottom corner of next slot

N100 G01 F30 X89 Y8 Z-10 M08; % Vertically plunging for milling

N105 G01 F30 Y52; % Move to top-left corner

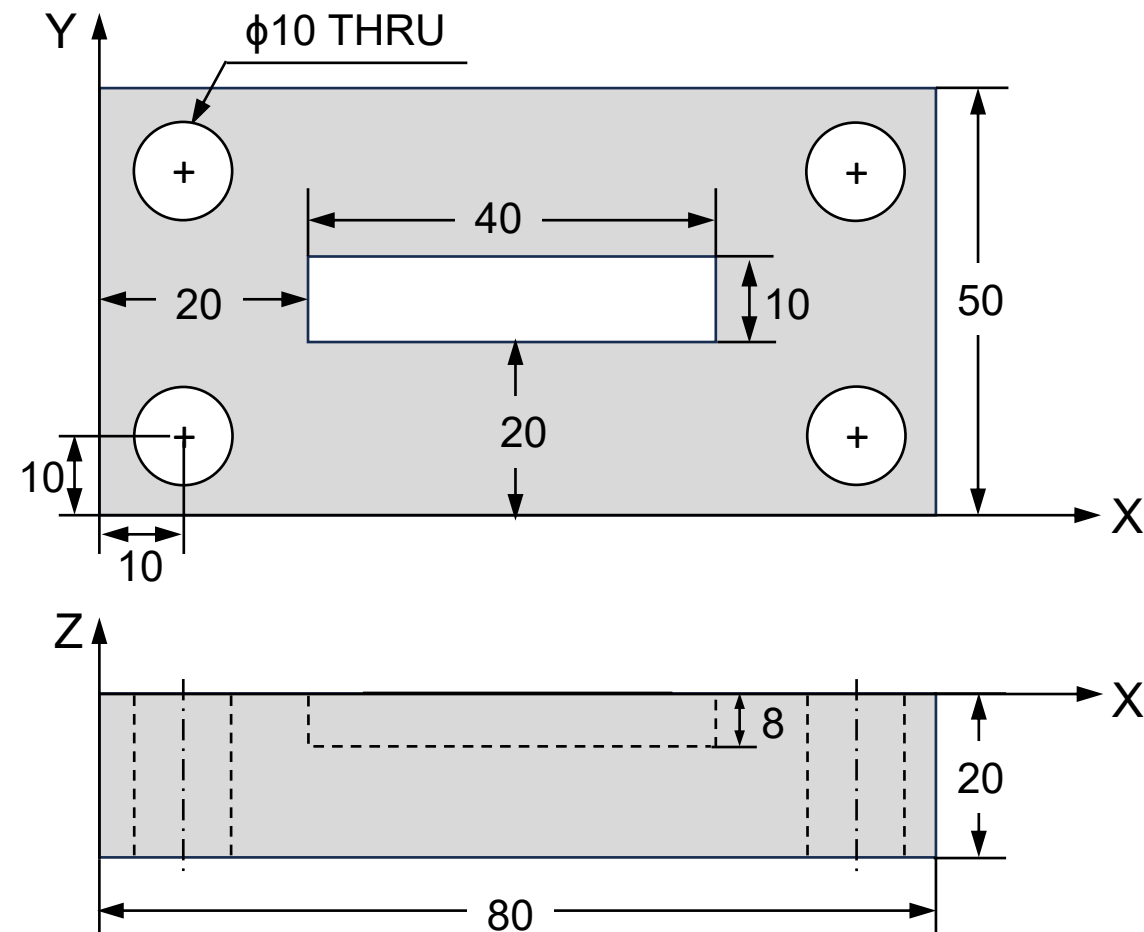
N110 G01 F30 X118; % Move to top-right corner

N115 G01 F30 Y8; % Move to bottom-right corner

N120 G01 F30 X89; % Move back to bottom-left corner



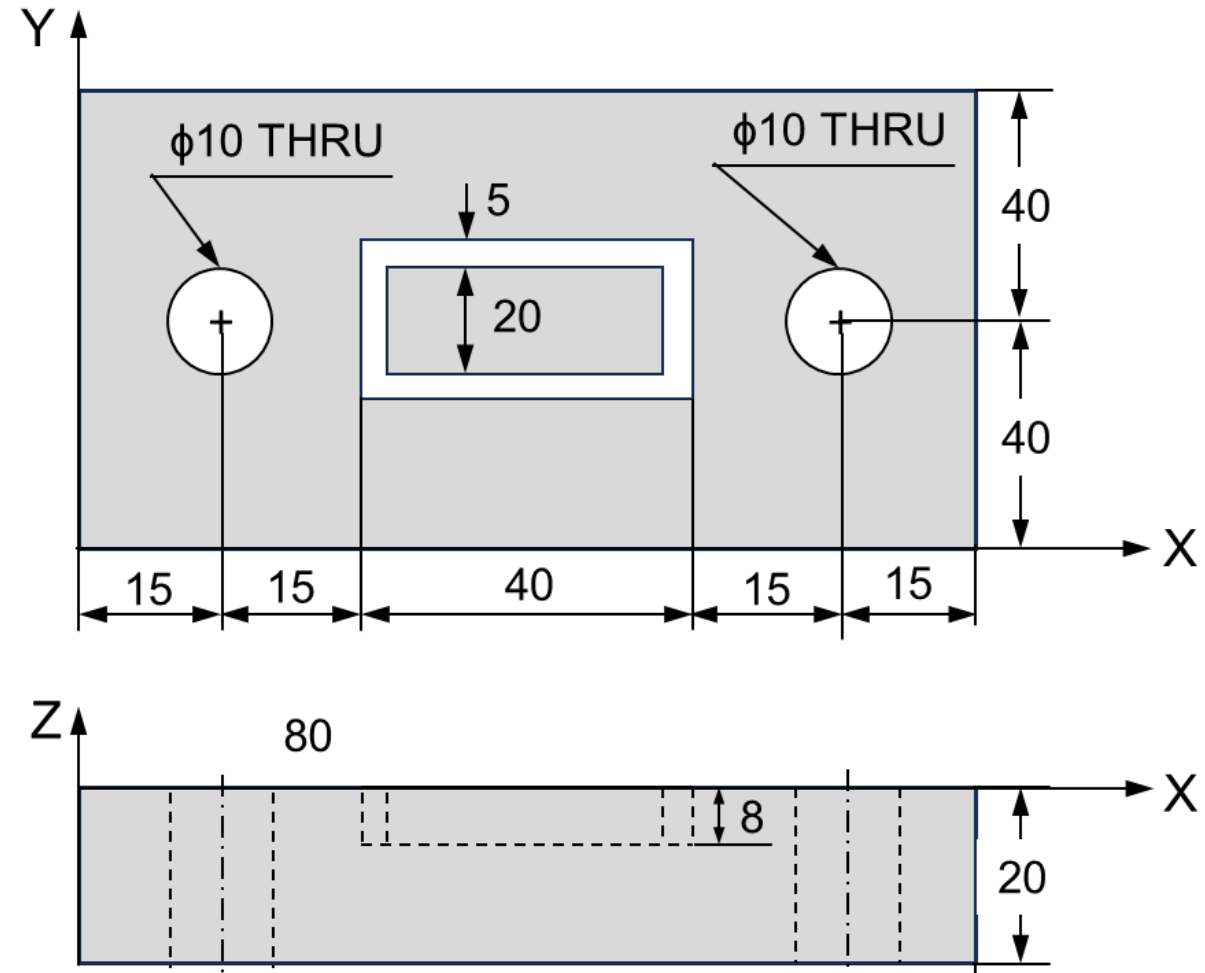
Question: Write a G&M code program for making a rectangular slot and four symmetric holes on a $80 \times 50 \times 20$ mm³ rectangular plate as shown below. Available end-milling tool (T015) has diameter of 10 mm and drill (T06) has diameter of 10 mm.



Do it yourself. Hint:

1. First attach the drill (T06)
2. Go to the location of first hole and drill.
3. Retract and go to the location of second hole, & drill.
4. Retract and go to the location of third hole, & drill.
5. Retract and go to the location of fourth hole, & drill.
6. Change the tool and attach the end-milling tool (T015)
7. Go to any one side of the slot and give depth
8. Move in X direction to cut the slot
9. Retract the tool and end the program

Question 1: Write a G&M code-based CNC program for drilling two 10-mm-diameter through holes and one rectangular-shaped blind channel within a $100 \times 80 \times 20$ mm³ rectangular plate, as per the adjacent schematic. Available drill (T01) has diameter of 10 mm. Available end-mill (T08) has 5 mm diameter. Consider appropriate rotational speed and feed velocity. Assume symmetry. Also assume appropriate value for any missing dimension or parameter. .





1. <https://youtu.be/2SLmVlfwtuU>
2. <https://youtu.be/oUBAT7eEwds>
3. <https://youtu.be/Fox4lBJwTuo>
4. <https://www.youtube.com/shorts/UBbL8ojg4wo>
5. https://www.youtube.com/shorts/zWvPF_cJMfE
6. <https://youtu.be/S8zDRu72HD0>
7. <https://youtu.be/wUruPB1xV0o>
8. <https://youtu.be/r0Ucw-0o5Mo>
9. <https://youtu.be/7Awl4h4K61c>